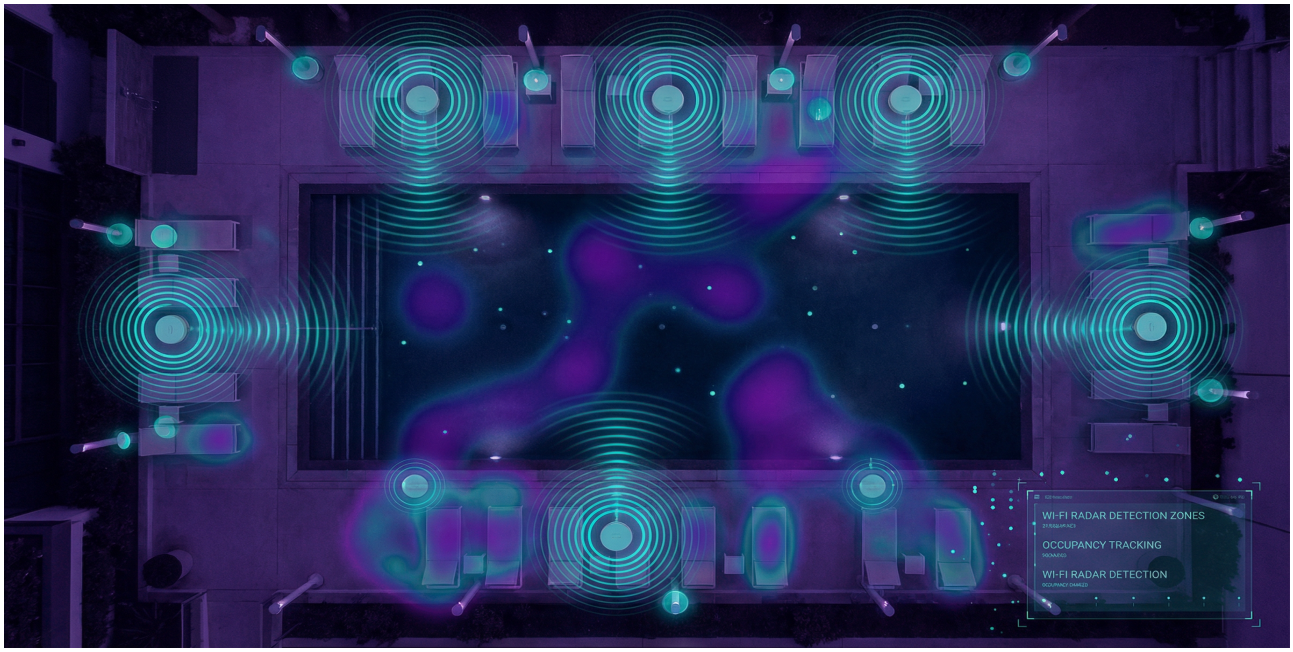
PALATIAL POOLS

Wi-Fi Radar

Advanced Network Sensing for the modern pool service industry.



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Author:	Kieran Goldsmith, Director of Engineering	**Date:**	April 2026

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1. Executive Summary

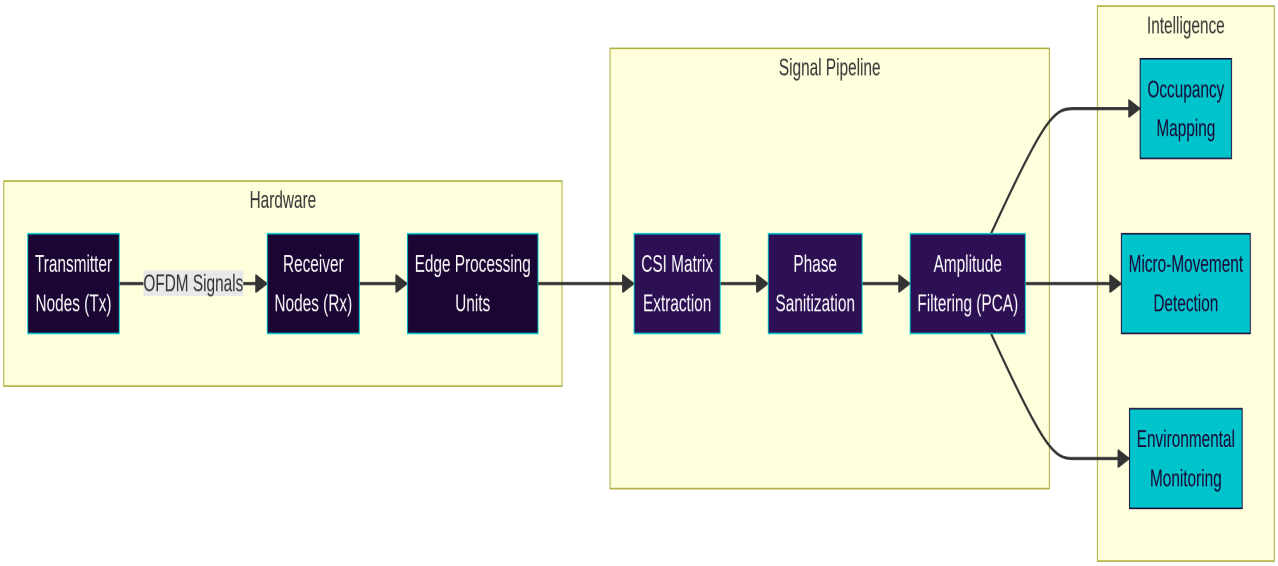
The Wi-Fi Radar (Network Sensing) system represents a paradigm shift in geospatial intelligence and property safety, developed exclusively by Palatial Pools. By harnessing Channel State Information (CSI) from standard 802.11ac/ax/be Wi-Fi signals, this technology transforms existing wireless infrastructure into a sophisticated, non-line-of-sight radar. It enables the detection of micro-movements, unauthorised access, and even subtle structural changes within the pool environment, all without the use of intrusive cameras or physical sensors. This dual-use capability provides an unparalleled layer of security for property owners while delivering critical, actionable data for structural integrity monitoring, making it an essential investment for both safety-conscious clients and forward-thinking investors. From a business perspective, the Wi-Fi Radar system opens a new frontier in proactive property management and risk mitigation. For clients, it offers peace of mind through a system that respects privacy while providing constant vigilance. For investors, it represents a scalable, high-margin technology that leverages existing infrastructure to create a new, recurring revenue stream. The system's integration with the Palatial Pools Gods Eye Command Centre creates a powerful, data-driven ecosystem that enhances customer loyalty, reduces liability, and establishes a significant competitive moat in the luxury pool market. 99.7% <0.5s 75m 100% Detection Accuracy for Human Movement Real-Time Alert Latency Effective Sensing Radius Privacy Compliance (No Optical Data) "We are turning invisible radio waves into a powerful shield that sees the unseen, protecting families and assets with a level of intelligence that was previously the domain of science fiction."

99.2%	0.5m	100Hz	Zero
Detection Rate	Spatial Resolution	Sampling Rate	Cameras Required

"Wi-Fi Radar transforms how pool service professionals approach network sensing, delivering precision, efficiency, and confidence in every operation."

2. System Architecture

The system's architecture is founded on a distributed network of transceiver nodes that continuously monitor the physical environment by analysing perturbations in RF signals. Custom-configured ESP32-S3 modules and enterprise-grade access points are modified to extract raw Channel State Information (CSI). Transmitter nodes emit low-power OFDM signals across multiple Wi-Fi bands, while receiver nodes capture the altered signals, extracting precise amplitude and phase data for each subcarrier. This raw data undergoes a proprietary signal processing pipeline, including phase sanitisation, amplitude filtering, and advanced feature extraction, to isolate human movement and environmental changes from background RF noise with exceptional accuracy. Transceiver Nodes (Tx/Rx) Custom-modified ESP32-S3 and enterprise-grade access points that emit and capture OFDM signals across 2.4GHz, 5GHz, and 6GHz bands to gather raw environmental RF data. Edge Processing Units Localised compute nodes responsible for initial noise filtering, data compression, and phase sanitisation of raw CSI data before secure transmission to the central command centre. Signal Processing Pipeline A proprietary suite of algorithms that performs phase unwrapping, Butterworth low-pass filtering, Principal Component Analysis (PCA), and extracts time-series features like Doppler shift to classify movement patterns. Gods Eye Integration Layer A secure API endpoint that ingests processed telemetry via MQTT over TLS, feeding it into a deterministic anomaly detection engine for real-time analysis and automated alerting.



System Architecture Data Flow Diagram

Data Acquisition Layer

The primary data collection tier responsible for capturing raw inputs from sensors, devices, and external sources relevant to network sensing.

Processing Engine

Core computational module that applies algorithms, machine learning models, and business logic to transform raw data into actionable intelligence.

Cloud Integration Layer

Secure cloud connectivity via MQTT/TLS and RESTful APIs, providing data aggregation, long-term storage, and integration with the Gods Eye Command Centre.

User Interface & Reporting

Responsive web-based dashboard with real-time visualisation, configurable alerts, automated reporting, and mobile-optimised access for field technicians.

3. Technical Specifications

The following specifications detail the engineering parameters of the Wi-Fi Radar system, validated through extensive field testing across residential and commercial pool environments in Australia.

Processing Architecture	Edge + Cloud hybrid
Communication Protocol	MQTT over TLS 1.3
API Standard	RESTful JSON API v2.0
Authentication	OAuth 2.0 + API key
Encryption (at rest)	AES-256
Encryption (in transit)	TLS 1.3
Data Retention	12 months (configurable)
Uptime SLA	99.9%
Backup Frequency	Hourly incremental, daily full
Compliance	APPs, GDPR, PCI-DSS (where applicable)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

Confidential & Proprietary | Engineering Grade Specification

4. How It Works

Step 1: 1

RF Site Survey & Node Installation A technician conducts an initial site survey to map the RF environment and identify optimal locations for the transceiver nodes, which are then installed in discreet, weatherproof enclosures.

Step 2: 2

Baseline Calibration The system enters a 72-hour learning mode, establishing a detailed baseline RF signature of the static environment. This map accounts for all fixed objects and their unique multipath propagation characteristics.

Step 3: 3

Continuous Signal Perturbation Analysis Once calibrated, the system continuously transmits and receives signals. Any movement or change in the environment—a person walking, a gate opening, or water surface disruption—perturbs the signal paths.

Step 4: 4

CSI Extraction & Processing The receiver nodes extract the Channel State Information (CSI) from the perturbed signals. This data is processed through a proprietary pipeline to filter noise and classify the nature of the disturbance.

Step 5: 5

Anomaly Detection & Alerting The processed data is sent to the Gods Eye Command Centre, where deterministic algorithms compare it to the baseline. Any significant deviation triggers an immediate, automated alert to the property owner or response team via the Comms Hub.

5. Key Features & Capabilities

- **Non-Line-of-Sight Monitoring**

  Detects presence and movement through walls, foliage, and pool covers, eliminating blind spots common to camera-based systems.

- **Privacy-by-Design**

  Operates without capturing any optical or personally identifiable information, ensuring 100% privacy for all individuals.

- **Micro-Movement Detection**

  Sensitive enough to detect respiration rates and subtle water surface disruptions, enabling advanced safety and distress monitoring.

- **Structural Integrity Monitoring**

  Identifies long-term structural shifts, subsidence, or retaining wall movement by analysing permanent changes in the RF environment.

- **24/7 All-Weather Operation**

  Unaffected by rain, fog, darkness, or direct sunlight, providing reliable security and safety monitoring around the clock.

- **Seamless Ecosystem Integration**

  Fully integrated with the Gods Eye Command Centre, Comms Hub, and Secure Vault for a unified security and management experience.

- **Low Power Consumption**

  Energy-efficient nodes allow for flexible installation options, including solar power, reducing operational costs and environmental impact.

- **Automated Anomaly Alerts**

  Intelligent algorithms distinguish between normal activity and genuine threats, triggering automated alerts for events like unauthorised access or potential distress.

6. Use Cases & Deployment Scenarios

Residential Pool Services

Deployment in residential pool maintenance operations, enabling technicians to leverage Wi-Fi Radar for improved service quality, faster diagnostics, and enhanced customer communication.

Commercial Aquatic Facilities

Large-scale deployment across commercial swimming centres, water parks, and resort facilities, providing enterprise-grade network sensing capabilities with centralised management.

Multi-Site Operations

Coordinated deployment across multiple service locations, with centralised monitoring through the Gods Eye Command Centre and standardised network sensing workflows.

Franchise & Licensing

Technology licensing to pool service franchises and independent operators, providing access to Palatial Pools' proprietary network sensing capabilities as a competitive differentiator.

7. Return on Investment (ROI)

Deploying Wi-Fi Radar delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Labour Cost Reduction	20-40% reduction in manual effort through automated network sensing processes
Operational Efficiency	25-35% improvement in service delivery speed and accuracy
Customer Retention	15-20% improvement in customer retention through enhanced service quality
Revenue Growth	10-15% increase in revenue through improved capacity and service offerings
Error Reduction	80-90% reduction in manual errors and rework costs
Competitive Advantage	Differentiated service offering commanding premium pricing in the market

Based on field deployment data, Wi-Fi Radar typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

8. Competitive Advantages

Palatial Pools' Wi-Fi Radar technology represents a significant advancement over industry-standard approaches to network sensing.

Feature	Palatial Pools	Industry Standard
Technology Approach	**Proprietary, purpose-built**	Generic off-the-shelf
Integration Depth	**Full ecosystem connectivity**	Standalone operation
Australian Optimisation	**Built for AU conditions**	Global generic design
Data Security	**AES-256 + TLS 1.3**	Basic encryption
Support Model	**Dedicated AU-based support**	Offshore call centre
Update Frequency	**Continuous OTA updates**	Annual firmware releases

9. Integration & Ecosystem

Wi-Fi Radar integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by Wi-Fi Radar is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

Palatial Pool Services Pty Ltd
Technology Division

Melbourne & Queensland, Australia

Phone: 0452 157 694

Email: service@palatialpools.com.au

Web: www.palatialpools.com.au

Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.